Counterparty Credit Risk Simulation

Full-book simplified SA-CCR with a Monte Carlo deep dive on CSA-005 (Fund2)

Diheng (Hector) Xin | M.S. Enterprise Risk Management | August 2026

Valuation date: 2025-06-15 | Source book: 120 trades | 8 netting sets | 6 counterparties | 5 asset classes

27.7% Netting benefit (full book)	**$478.5M** Simplified SA-CCR EAD (full book)	**$9.77M** EPE, CSA-005 (uncollateralized)
91.9% EPE reduction from collateral	**$142,759** CVA, CSA-005 (collateralized)	**+0.6%** Within-model WWR add-on to CVA

Project scope: Parts 1-2 provide a portfolio-wide regulatory-style view. Parts 3-6 focus on one representative netting set for detailed exposure simulation, collateral, CVA, and wrong-way-risk analysis. This illustrates a practical CCR workflow while keeping the deeper modeling exercise focused and transparent.

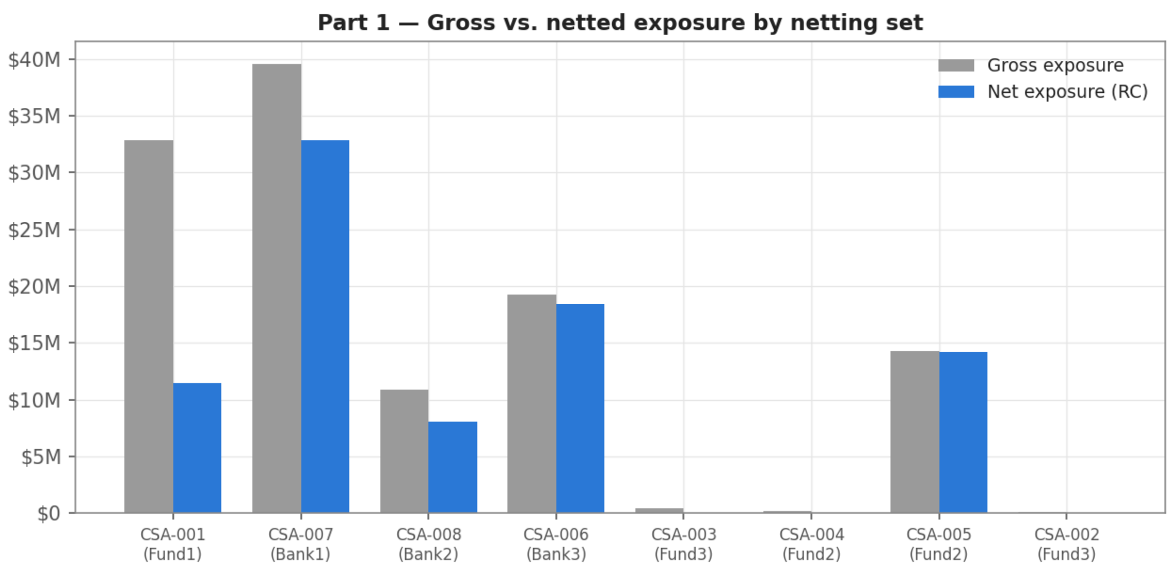
All data, counterparties, trades, CSA terms, and results are synthetic and anonymized. This report is for educational and recruiting purposes only.

Part 1 - Netting

Trades are grouped by netting agreement under an assumed legally enforceable ISDA Master Agreement. Positive and negative MTM values offset within each set before exposure is floored at zero.

Counterparty	Netting Set	Gross Exposure	Net Exposure (RC)	Benefit	Benefit %
Fund1	CSA-001	$32,840,763	$11,445,044	$21,395,719	65.1%
Bank1	CSA-007	$39,576,883	$32,865,173	$6,711,711	17.0%
Bank2	CSA-008	$10,907,787	$8,041,649	$2,866,138	26.3%
Bank3	CSA-006	$19,252,018	$18,409,494	$842,524	4.4%
Fund3	CSA-003	$453,525	$0	$453,525	100.0%
Fund2	CSA-004	$166,810	$0	$166,810	100.0%
Fund2	CSA-005	$14,256,511	$14,192,282	$64,229	0.5%
Fund3	CSA-002	$58,068	$0	$58,068	100.0%

Portfolio-wide netting benefit: 27.7%. Gross exposure of $117.5M nets down to $85.0M of replacement cost. A 100% benefit means the net MTM is negative and replacement cost is therefore floored at zero.

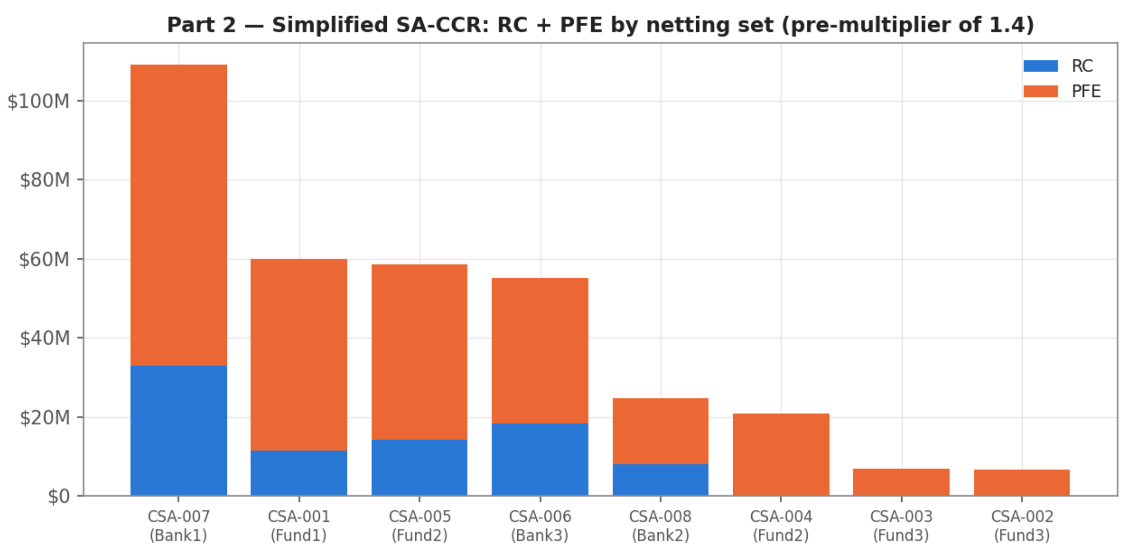


Part 2 - Simplified SA-CCR EAD

The full-book calculation applies supervisory delta, supervisory duration, maturity factors, hedging-set aggregation, the PFE multiplier, and EAD = 1.4 x (RC + PFE).

Counterparty	Netting Set	RC	Total Add-On	Multiplier	PFE	EAD
Bank1	CSA-007	$32,865,173	$76,210,207	1.00	$76,210,207	$152,705,531
Fund1	CSA-001	$11,445,044	$48,569,604	1.00	$48,569,604	$84,020,507
Fund2	CSA-005	$14,192,282	$44,424,920	1.00	$44,424,920	$82,064,083
Bank3	CSA-006	$18,409,494	$36,609,514	1.00	$36,609,514	$77,026,611
Bank2	CSA-008	$8,041,649	$16,706,526	1.00	$16,706,526	$34,647,445
Fund2	CSA-004	$0	$24,091,241	0.87	$20,877,284	$29,228,198
Fund3	CSA-003	$0	$7,542,342	0.90	$6,773,905	$9,483,467
Fund3	CSA-002	$0	$9,054,306	0.74	$6,694,372	$9,372,120

Total simplified EAD: $478.5M. The PFE multiplier is below 1.0 for netting sets with negative net MTM, recognizing that negative current exposure reduces - but does not eliminate - potential future exposure.



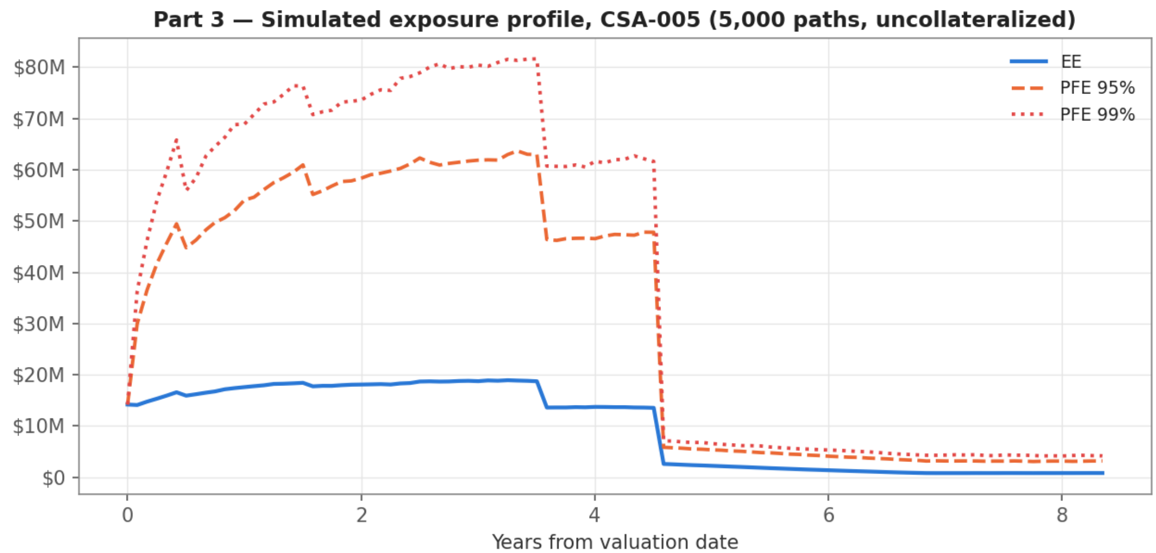
This section is deliberately labeled simplified: rate maturity-bucket correlations, collateralized replacement cost, and Basel-prescribed option volatilities are not fully implemented.

Part 3 - Monte Carlo Exposure Simulation

The detailed analysis focuses on CSA-005 (Fund2). The model simulates 5,000 correlated monthly paths over 8.35 years using GBM for Equity and FX and log-space mean reversion for Rates, Commodity, and Credit. Cholesky decomposition applies the assumed correlation structure.

Eight live trades are revalued with product-sensitive simplified methods: forward-style valuation for Equity, FX, and Commodity; annuity-scaled valuation for Rates and CDS; and a calibrated Black-76-style treatment for the swaption.

Years	Date	EE	PFE 95%	PFE 99%
0.0	2025-06-15	$14,192,282	$14,192,282	$14,192,282
1.0	2026-06-15	$17,591,629	$54,092,390	$68,898,863
2.0	2027-06-15	$18,094,523	$58,362,980	$73,594,664
3.0	2028-06-15	$18,749,802	$61,877,674	$80,419,647
5.0	2030-06-15	$2,233,693	$5,349,694	$6,556,058
8.0	2033-06-15	$820,896	$3,159,164	$4,154,660

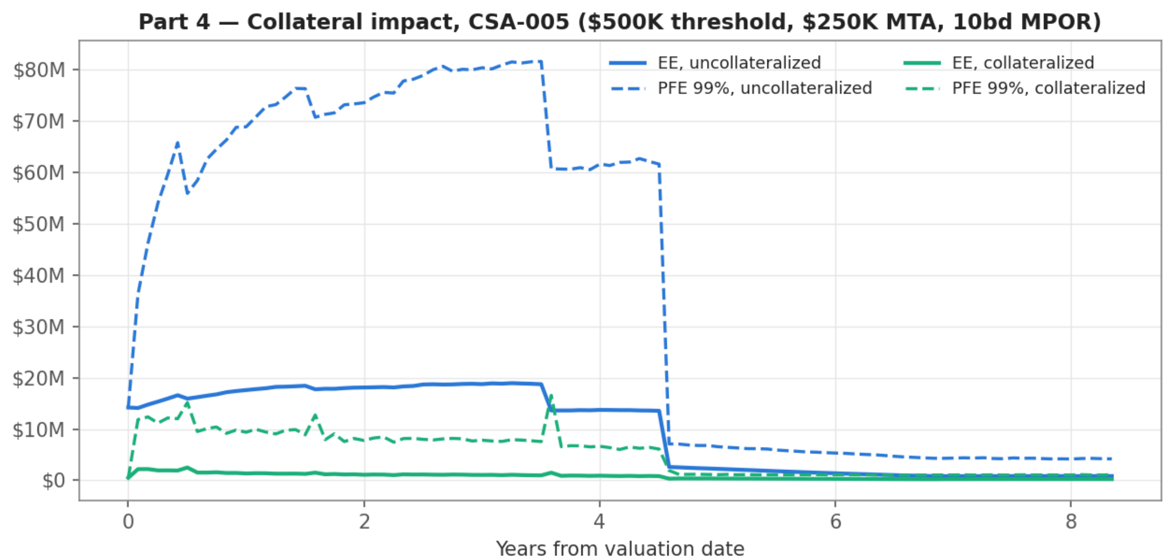


Lifetime EPE: $9,769,587. Exposure rises through the early years as market uncertainty grows, then declines in stages as trades mature. A major decline occurs when the largest CDS position rolls off.

Part 4 - Collateral and CSA Modeling

Collateral is modeled using a $500K threshold, $250K minimum transfer amount, and 10-business-day margin period of risk. A daily collateral approximation is built from linearly interpolated monthly exposures. Held collateral changes only when the transfer amount reaches the MTA and is evaluated with the MPOR lookback.

Years	EE Before	EE After	PFE 99% Before	PFE 99% After
0.0	$14,192,282	$500,000	$14,192,282	$500,000
1.0	$17,591,629	$1,338,004	$68,898,863	$9,377,584
2.0	$18,094,523	$1,083,400	$73,594,664	$7,771,428
3.0	$18,749,802	$1,045,991	$80,419,647	$7,818,789
5.0	$2,233,693	$347,458	$6,556,058	$1,104,224
8.0	$820,896	$235,057	$4,154,660	$1,048,051

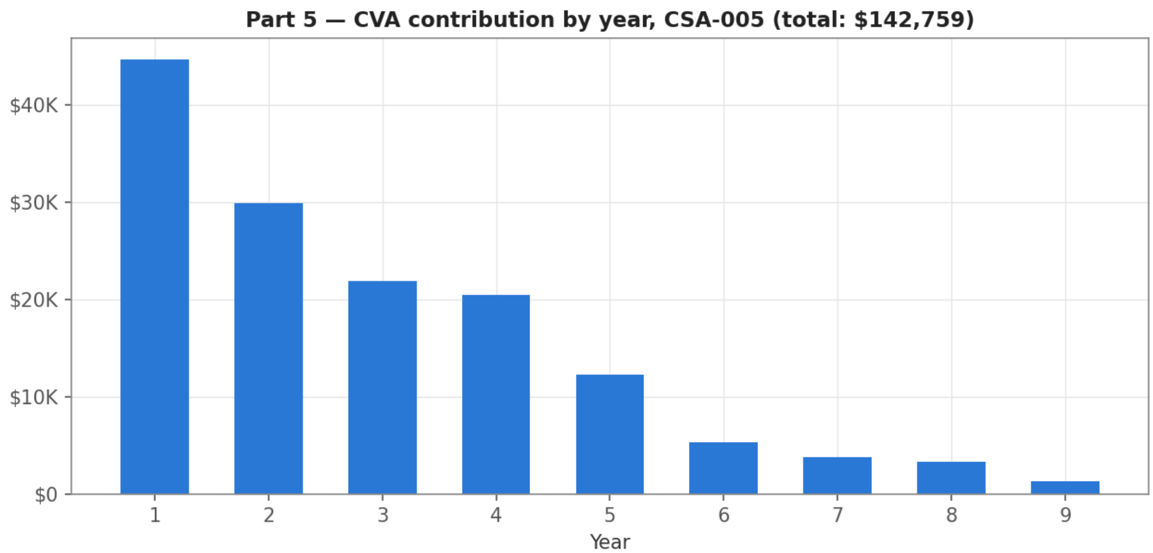


EPE falls 91.9%, from $9.77M to $794,518. PFE remains above the threshold because the MTA and MPOR create residual exposure gaps. This residual exposure is why CVA can remain material on a collateralized portfolio.

Part 5 - Credit Valuation Adjustment

CVA is calculated from the collateralized EE profile. The counterparty credit-spread proxy is the median of CSA-005's live-trade spreads. A 60% LGD converts the spread to a flat hazard rate, and future expected losses are discounted at a flat 4% risk-free rate.

Metric	Value
Credit spread (median of live trades)	264.9 bps
Credit spread sensitivity range (25th-75th percentile)	82.1 - 353.7 bps
LGD assumption	60%
Implied flat hazard rate	4.41%
Cumulative default probability (8.35y)	30.8%
CVA, uncollateralized (context)	$1,769,585
CVA, collateralized	$142,759
CVA sensitivity range	$47,551 - $184,333



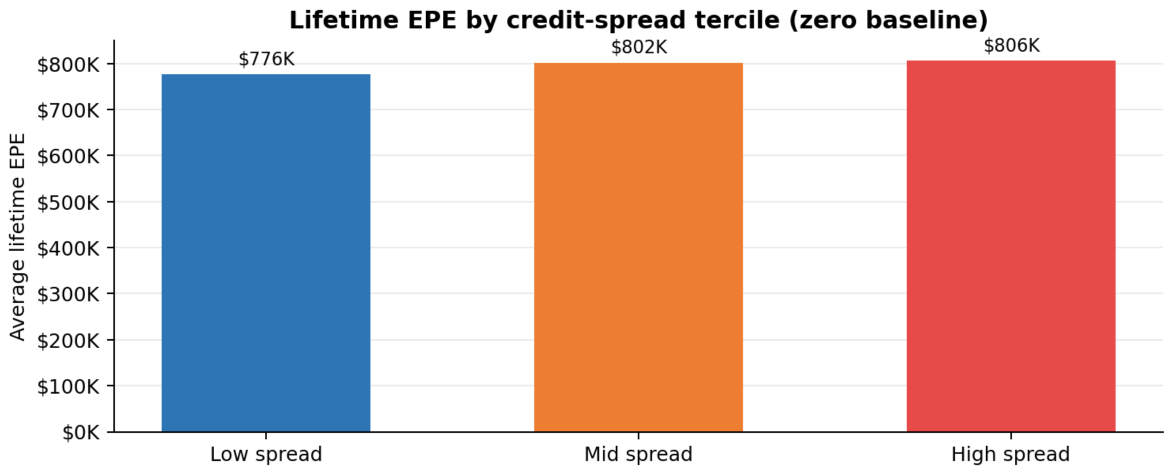
Collateralized CVA: $142,759. CVA is front-loaded: Year 1 contributes approximately 31% of the total, then declines as exposure rolls off and marginal default probability decreases.

Part 6 - Wrong-Way Risk

Fund2's credit-spread proxy is linked to the simulated ITRAXX.EU path. Joint CVA preserves the path-by-path relationship between exposure and default risk. The independence benchmark uses the same stochastic spread distribution but separates it from exposure, isolating the within-model dependence effect.

Metric	Value
CVA, independent benchmark	$145,335
CVA, joint path-linked result	$146,252
Within-model WWR add-on	$916 (+0.6%)
Scenario classification	Modest wrong-way-risk signal
Correlation: path spread vs. path EPE	+0.041
WWR add-on, uncollateralized (context)	+1.4%

Tercile	Average Spread	Average Lifetime EPE	Paths
Low spread	223.9 bps	$775,916	1,667
Mid spread	269.0 bps	$801,571	1,666
High spread	326.3 bps	$806,072	1,667



The scenario produces a modest positive WWR signal: average lifetime EPE increases across spread terciles, and the joint CVA exceeds the independence benchmark by 0.6%. Collateral reduces the effect from 1.4% before collateral to 0.6% afterward.

Limitations and Interpretation

This project is designed to demonstrate CCR methodology and model judgment, not to reproduce a production pricing or regulatory-capital platform.

- Simplified SA-CCR: no rate maturity-bucket correlation, no collateralized replacement cost, and trade-level rather than Basel-prescribed option volatilities.
- Focused simulation scope: Parts 3-6 cover CSA-005 only, not the full book or Fund2's complete relationship.
- Stylized market models: dynamics, volatilities, and correlations are scenario assumptions rather than calibrated market inputs.
- Simplified pricing: product-sensitive approximations are used instead of full curves, cash-flow schedules, and production valuation models.
- CVA assumptions: the hazard rate and discount rate are flat because the synthetic dataset does not contain market curves.
- WWR proxy: the counterparty spread is linked to a broad credit index rather than a calibrated single-name credit process.
- Collateral scope: the model includes Threshold, MTA, and MPOR but excludes initial margin, haircuts, collateral interest, and dispute periods.
- Data quality: known inconsistencies in CSA terms and credit spreads are handled through explicit scenario assumptions and sensitivity analysis.

Selected methodology references

Basel Committee on Banking Supervision, Basel Framework: CRE52 - Standardised Approach to Counterparty Credit Risk.

Standard market relationships used in the project include the credit triangle approximation, exponential survival probabilities, discounted expected-loss CVA, and Monte Carlo exposure simulation.

Data disclaimer

All counterparties, trades, exposures, CSA terms, agreement identifiers, and results are fictional and anonymized. Any resemblance to an actual institution or portfolio is coincidental.